

Exp.No 5 IOC-Based Detection

Aim:-

To detect malicious activity by:

- Reading an IOC list (malicious IPs / file hashes)
- Parsing log files
- Comparing logs against IOC indicators
- Flagging suspicious events automatically using Python

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NameError: name 'cat' is not defined

(unknown@kali) ~/IOCS
$ python3 ioc_detection.py
=====
IOC BASED DETECTION SYSTEM
=====

IOC List Loaded Successfully
type      value
0 IP      45.33.32.156
1 IP      103.21.244.0
2 HASH    5f4dcc3b5aa765d61d8327deb882cf99
3 HASH    098f6bcd4621d373cade4e832627b4f6
4 EOF     NaN

Log File Loaded Successfully
timestamp source_ip destination_ip file_hash action
0 10:01 192.168.1.10 45.33.32.156 abc123 Allowed
1 10:05 192.168.1.20 8.8.8.8 5f4dcc3b5aa765d61d8327deb882cf99 Blocked
2 10:10 192.168.1.15 103.21.244.0 xyz789 Allowed
3 10:15 192.168.1.25 1.1.1.1 testhash Allowed
4 10:20 192.168.1.30 9.9.9.9 098f6bcd4621d373cade4e832627b4f6 Blocked
5 EOF NaN NaN NaN NaN

Detection Results:

timestamp source_ip destination_ip file_hash action IP_Match Hash_Match IOC_Flag
0 10:01 192.168.1.10 45.33.32.156 abc123 Allowed True False True
1 10:05 192.168.1.20 8.8.8.8 5f4dcc3b5aa765d61d8327deb882cf99 Blocked False True True
2 10:10 192.168.1.15 103.21.244.0 xyz789 Allowed True False True
3 10:15 192.168.1.25 1.1.1.1 testhash Allowed False False False
4 10:20 192.168.1.30 9.9.9.9 098f6bcd4621d373cade4e832627b4f6 Blocked False True True
5 EOF NaN NaN NaN NaN False False False

Suspicious Events Detected:

timestamp source_ip destination_ip file_hash action IP_Match Hash_Match IOC_Flag
0 10:01 192.168.1.10 45.33.32.156 abc123 Allowed True False True
1 10:05 192.168.1.20 8.8.8.8 5f4dcc3b5aa765d61d8327deb882cf99 Blocked False True True
2 10:10 192.168.1.15 103.21.244.0 xyz789 Allowed True False True
4 10:20 192.168.1.30 9.9.9.9 098f6bcd4621d373cade4e832627b4f6 Blocked False True True

Detection Completed Successfully.
Traceback (most recent call last):
  File "/home/unknown/IOCS/ioc_detection.py", line 36, in <module>
    EOF
NameError: name 'EOF' is not defined

(unknown@kali) ~/IOCS
$
```